Ayush Ghatak

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Profile Summary

M.Sc. Data Science graduate with hands-on research and project experience spanning LLM fine-tuning, computer vision, predictive modeling, and large-scale decision systems, with experience in model deployment and inference optimization.

Technical Skills

Languages: Python, SQL, R
AI/ML: Scikit-learn, XGBoost, TensorFlow, Transformers, Hugging Face, PEFT, OpenCV, NetworkX, PyTorch
Libraries & Tools: NumPy, Pandas, SciPy, Gradio

Experience

AI Research Intern, IAI, TCG Crest — *June 2025 – April 2026*

- Built a Kidney Exchange optimization system on UNOS-like data for 10K+ patients, cutting runtime by $\approx$ 13% over prior approaches.
- Implemented heuristic matching algorithms for large-scale donor-patient allocation, balancing efficiency and fairness constraints across varied pool configurations.
- Trained an XGBoost model to predict allocation outcomes, achieving MAE of 2.45% and R^2 of 0.90.
- Built FUTUREMATCH, a forecasting framework that scores donor-patient pairs to prioritize high-value matches in dynamic exchange pools.

Major Projects

Legal Clause Simplification using Knowledge Distillation — *Github ↗*
Tech Stack: Python, PEFT, Transformers, Hugging Face, Gradio, GGUF

- Built an end-to-end Legal AI application that transforms complex legal clauses into plain explanations using LLM fine-tuning and Chain-of-Thought knowledge distillation.
- Fine-tuned *Qwen-2.5-3b-instruct* with QLoRA on a domain-specific corpus generated by *Llama-4-scout-17b-16e-instruct* as a teacher model, improving legal reasoning and explanation quality.
- Integrated zero-shot legal-domain classification to filter non-legal inputs and deployed the model on Hugging Face Spaces with a Gradio interface, enabling CPU-only inference through 5-bit GGUF optimization.

Drone-Based 3D Reconstruction Optimization — *Github ↗*
Tech Stack: Python, OpenCV, NumPy, SciPy, PyProj, Computer Vision, Numerical Optimization

- Built a data-driven 3D reconstruction pipeline on 176 high-resolution drone images using SIFT feature extraction and statistical filtering techniques including Lowe's ratio test and RANSAC.
- Formulated reprojection error as an optimization objective, reducing reconstruction error by 84% (18.71 to 2.91 pixels) via BFGS, and improved spatial alignment by 94.6% through geospatial transformation using Newton's method.
- Achieved a final reconstruction error of 0.23 m through iterative cost minimization using Fibonacci search.

HospitalFlow AI: Predictive Modeling & Patient Segmentation — *Github ↗*
Tech Stack: Python, Scikit-learn, XGBoost, TensorFlow/Keras, NumPy, Pandas

- Built an end-to-end ML pipeline on 100K+ healthcare records using feature engineering, predictive modeling using Linear Regression, XGBoost, and Neural Networks for Length-of-Stay prediction (MAE: 1.84 days).
- Applied Isolation Forest and K-Means++ clustering ($K = 6$) on 99K+ encounters to identify high-risk patient cohorts and clinically meaningful patient clusters.

Education

M.Sc. Data Science, St. Xavier's College (Autonomous), Kolkata — CGPA: 9.02 — *2024-2026*
B.Sc. Computer Science, St. Xavier's College (Autonomous), Kolkata — CGPA: 8.73 — *2021-2024*
Class XII (CBSE), Notre Dame School, Delhi — 94.4% — *2021*

Achievements & Activities

- Secured first place in PowerQuest Hackathon 2025 (HEOR Analytics), PharmaQuant Insights Pvt. Ltd.